

AI Voice Assistant — Project Plan

Overview

This project aims to build an **AI Voice Assistant** that allows users to interact completely through voice — speaking a query and receiving a spoken response, similar to how a human conversation works.

The system will be built in a **phase-wise manner**, starting with a simple text-based chatbot and gradually evolving into a full voice-based conversational assistant. Each component of the system is designed to be **modular**, meaning individual parts (like the speech recognition model or the voice output model) can be replaced or upgraded independently without affecting the rest of the system.

Objective

To design and develop a voice assistant capable of:

- Understanding spoken user queries
- Processing them intelligently using an AI language model
- Responding back in natural, human-like speech
- Supporting real-time, low-latency conversation in later stages

Final Target Flow

User Voice → Speech-to-Text (STT) → Gemini API (LLM) → Text Response → Text-to-Speech (TTS) → Spoken Output

Why a Phase-Wise Approach?

Building a real-time voice assistant directly is complex, since it involves multiple AI components working together (speech recognition, language understanding, and speech generation). To reduce risk and make debugging easier, the project is broken into smaller phases — each phase builds on the previous one and is tested independently before moving forward.

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Project Phases

Phase 1: Text-Based Chatbot (Foundation)

Build a chatbot that can understand and respond to text queries using the Gemini API.

- User types a query
- The query is sent to the Gemini API
- The API generates a response
- The response is displayed as text

Goal: Establish the core "brain" of the assistant and confirm that conversation logic (remembering context, generating replies) works correctly before adding audio.

Phase 2: Adding Speech-to-Text (Voice Input)

Enable the assistant to accept voice input instead of typed text.

- User speaks into a microphone
- The audio is converted into text using a Speech-to-Text (STT) model/API
- The transcribed text is passed to the same chatbot logic built in Phase 1

Goal: Allow the assistant to "hear" the user.

Phase 3: Adding Text-to-Speech (Voice Output)

Enable the assistant to reply using natural spoken voice instead of only text.

- The text response generated by Gemini is converted into audio using a Text-to-Speech (TTS) model/API
- The audio is played back to the user

Goal: Allow the assistant to "speak" back, completing the full voice-in, voice-out loop.

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System Design Principle: Modularity

Each core function of the assistant — Speech-to-Text, Language Processing, and Text-to-Speech — is treated as an **independent, swappable module**. This is done by defining a fixed set of rules (interfaces) for how each module receives input and gives output, regardless of which underlying technology or API is used.

Benefit: If a better STT or TTS technology becomes available in the future, it can be swapped in by changing only that module, without rewriting the rest of the assistant.

Technology Components (Planned)

Component	Purpose	Example Tools/APIs
Language Model (LLM)	Understands queries and generates responses	Gemini API
Speech-to-Text (STT)	Converts spoken voice into text	Whisper / Gemini Audio Input
Text-to-Speech (TTS)	Converts text responses into natural speech	ElevenLabs / Google TTS
Backend	Connects all components together	Python, FastAPI
Frontend	User interface for interaction	Web-based (HTML/JS or React)

Expected Outcome

By the end of the project, the system will function as a complete voice assistant where a user can speak naturally, and the assistant will understand, think, and reply back in a human-like spoken voice.